

E-Commerce Sales Analysis Using SQL Server

Project Overview

This project presents a comprehensive analysis of an e-commerce sales dataset using Microsoft SQL Server and SQL Server Management Studio (SSMS). The objective was to leverage SQL for data cleaning, transformation, aggregation, and reporting to uncover actionable business insights from retail transaction data.

Database Setup and Data Preparation

The dataset was imported into SQL Server, where several data preparation tasks were performed to ensure data quality and consistency. These included:

- Importing the dataset into SQL Server Management Studio (SSMS).
- Validating data integrity and identifying missing values.
- Cleaning and standardizing data fields.
- Creating derived columns such as Order Year and Order Month for time-based analysis.
- Executing SQL queries to aggregate, filter, and summarize key business metrics.

Business Insights

Overall Performance Metrics

The analysis produced the following key performance indicators (KPIs):

- Total Orders: 1,200
- Total Revenue: \$1,264,761.96
- Average Order Value: \$1,053.97
- Average Quantity per Order: 2 Items
- Cancelled Orders: 250

Customer and Payment Insights

Customer purchasing behavior revealed several notable trends:

- Most Frequently Used Payment Method: Online Payments (258 Transactions)
- Highest Revenue Payment Method: Credit Card (\$263,847.63)
- **Top Referral Source by Customer Volume:** Instagram (260 Referrals)
- **Highest Revenue Referral Source:** Instagram (\$275,285.45)

Product Performance

Product-level analysis identified the highest revenue-generating items across the dataset.

Top Five Products by Revenue

Rank Product Revenue

1	Chair	\$195,620.11
2	Printer	\$195,612.61
3	Laptop	\$192,126.56
4	Tablet	\$186,568.95
5	Monitor	\$175,651.41

Additional products generating more than \$150,000 in revenue included Desk and Phone.

Highest-Value Transaction

- Product: Tablet
- Coupon Code: SAVE10
- Transaction Value: \$3,456.40

Lowest-Value Transaction

- Product: Phone
- Referral Source: Email
- Transaction Value: \$11.39

Revenue Analysis

Revenue by Payment Method

Payment Method	Revenue
Credit Card	\$263,847.63
Online Payment	\$262,442.94
Cash	\$259,786.29
Gift Card	\$246,323.92
Debit Card	\$232,361.18

Revenue by Referral Source

Referral Source	Revenue
Instagram	\$275,285.45
Email	\$261,808.55
Google	\$250,441.48
Facebook	\$250,410.90
Referral	\$226,815.58

Key Findings

- Credit Card transactions generated the highest revenue despite Online Payments being the most frequently used payment method.

- Instagram emerged as both the leading customer acquisition channel and the highest revenue-generating referral source.
- Chairs, Printers, and Laptops were the strongest-performing products based on total revenue.
- Revenue distribution across payment methods was relatively balanced, indicating customers actively used multiple payment options.
- A small number of high-value transactions contributed significantly to overall sales performance.

Conclusion

This project demonstrates practical proficiency in SQL for data cleaning, querying, aggregation, and business reporting. By transforming raw transactional data into meaningful insights, the analysis provides valuable information on customer behavior, product performance, payment preferences, referral effectiveness, and overall revenue trends.

The project highlights the ability to apply SQL to solve real-world business problems and deliver data-driven insights that support strategic decision-making.